

Displays

D1:

$$\partial_t x = f(x), \quad x(t_0) \in \mathcal{B}_0 = B(x_0, \delta_0), \quad (1)$$

D2:

$$\delta_j \geq \max_{x \in \mathcal{B}_0} \|\chi(t_j, x) - \chi(t_j, x_0)\| = \max_{x \in \mathcal{B}_0} d(t_j, x), \quad (2)$$

D3:

$$\Delta \lambda_{\mathcal{V}}(\gamma) = F_{L, \hat{\gamma}}^{-1}(\sqrt{1 - \gamma}), \quad (3)$$

D4:

$$r_x = \frac{\left(-\lambda_x + \sqrt{\lambda_x^2 + 4 \cdot \Delta \lambda_{x, \mathcal{V}} \cdot (\mu \cdot \bar{m}_{j, \mathcal{V}} - d_j(x))}\right)}{2 \cdot \Delta \lambda_{x, \mathcal{V}}}, \quad (4)$$

D5:

$$\Pr(d_j(y) \leq \mu \cdot \bar{m}_{j, \mathcal{V}}) \geq 1 - \gamma \quad \forall y \in B(x, r_x)^S, \quad (5)$$

D6:

$$\forall \gamma \in (0, 1), \exists N \in \mathbb{N} \text{ s.t. } \Pr(\mu \cdot \bar{m}_{j, \mathcal{V}} \geq m_j^*) \geq 1 - \gamma \quad (6)$$

D7:

$$D_n^- = \sup_x (G_n(x) - \hat{F}_n(x)) \quad (S1)$$

D8:

$$\epsilon_{n, \gamma} = \sqrt{\frac{\ln \frac{1}{\alpha}}{2n}} \quad (S2)$$

D9:

$$F_{L, \gamma}(x) = G_n(x) - \epsilon_{n, \gamma} - D_n^- \quad (S3)$$

D10:

$$\Pr(\sup_x (F_{L, \gamma}(x) - F(x)) \leq 0) \geq 1 - \gamma, \quad (S4)$$

D11:

$$\Pr(\sup_x (\hat{F}_n(x) - F(x)) > \epsilon) \leq e^{-2n\epsilon^2} \quad (S5)$$

D12:

$$\Pr(\sup_x (\hat{F}_n(x) - F(x)) > \epsilon_{n, \gamma}) \leq \gamma, \quad (S6)$$

D13:

$$\Pr\left(\sup_x (G_n(x) - F(x)) > \epsilon_{n, \gamma} + D_n^-\right) = \quad (S7)$$

D14:

$$= \Pr\left(\sup_x (G_n(x) - \hat{F}_n(x) + \right. \quad (S8)$$

D15:

$$\left. + \hat{F}_n(x) - F(x)) > \epsilon_{n, \gamma} + D_n^-\right) \quad (S9)$$

D16:

$$\leq \Pr\left(\sup_x (G_n(x) - \hat{F}_n(x) + \right. \quad (S10)$$

D17:

$$\left. + \sup_x (\hat{F}_n(x) - F(x)) > \epsilon_{n, \gamma} + D_n^-\right) \quad (S11)$$

D18:

$$\stackrel{(S1)}{=} \Pr\left(\sup_x (\hat{F}_n(x) - F(x)) > \epsilon_{n, \gamma}\right) \quad (S12)$$

D19:

$$\stackrel{(S6)}{\leq} \gamma, \quad (S13)$$

D20:

$$\Delta\lambda_{\mathcal{V}}(\gamma) = F_{L,\hat{\gamma}}^{-1}(\sqrt{1-\gamma}), \quad (\text{S14})$$

D21:

$$r_x = \frac{\left(-\lambda_x + \sqrt{\lambda_x^2 + 4 \cdot \Delta\lambda_{x,\mathcal{V}} \cdot (\mu \cdot \bar{m}_{j,\mathcal{V}} - d_j(x))}\right)}{2 \cdot \Delta\lambda_{x,\mathcal{V}}}, \quad (\text{S15})$$

D22:

$$\Pr(d_j(y) \leq \mu \cdot \bar{m}_{j,\mathcal{V}}) \geq 1 - \gamma \quad \forall y \in B(x, r_x)^S, \quad (\text{S16})$$

D23:

$$\Pr(F_{L,\hat{\gamma}(x)} \leq F(x)) \geq 1 - \hat{\gamma} = \sqrt{1-\gamma} \quad (\text{S17})$$

D24:

$$\Pr(X \leq \Delta\lambda_{\mathcal{V}}) \geq \quad (\text{S18})$$

D25:

$$\geq \Pr(X \leq \Delta\lambda_{\mathcal{V}} | F_{L,\hat{\gamma}(x)} \leq F(x)). \quad (\text{S19})$$

D26:

$$\cdot \Pr(F_{L,\hat{\gamma}(x)} \leq F(x)) \quad (\text{S20})$$

D27:

$$\Pr(X \leq \Delta\lambda_{\mathcal{V}}) \geq F_{L,\hat{\gamma}}(\Delta\lambda_{\mathcal{V}}) \cdot \sqrt{1-\gamma} \quad (\text{S21})$$

D28:

$$\begin{aligned} \Pr(X \leq \Delta\lambda_{\mathcal{V}}) &\geq 1 - \gamma, \\ \text{with } X &= \max_{i=1}^m \left[\frac{|\lambda_{a_i} - \lambda_{b_i}|}{\|a_i - b_i\|} \right] \end{aligned} \quad (\text{S22})$$

D29:

$$\Pr(Y \leq \Delta\lambda_{\mathcal{V}}) \geq \Pr(X \leq \Delta\lambda_{\mathcal{V}}) \geq 1 - \gamma \quad (\text{S23})$$

D30:

$$\begin{aligned} \lambda_y &= \lambda_x + \frac{\lambda_y - \lambda_x}{\|x - y\|} \cdot \|x - y\| \\ &\leq \lambda_x + \frac{|\lambda_x - \lambda_y|}{\|x - y\|} \cdot \|x - y\| \end{aligned} \quad (\text{S24})$$

D31:

$$\Pr\left(\lambda_x + \frac{|\lambda_x - \lambda_y|}{\|x - y\|} \cdot \|x - y\| \leq \quad (\text{S25})$$

D32:

$$\leq \lambda_x + \Delta\lambda_{x,\mathcal{V}} \cdot \|x - y\| \Big) \geq 1 - \gamma \quad (\text{S26})$$

D33:

$$\Pr(\lambda_y \leq \lambda_x + \Delta\lambda_{x,\mathcal{V}} \cdot \|x - y\|) \geq 1 - \gamma \quad (\text{S27})$$

D34:

$$\begin{aligned} |d_j(x) - d_j(y)| &= |\|\chi(t_j, x) - \chi(t_j, x_0)\| - \\ &\quad - \|\chi(t_j, y) - \chi(t_j, x_0)\|| \quad \{\text{triangle inequality}\} \\ &\leq \|\chi(t_j, x) - \chi(t_j, y)\| \quad \{\text{mean value theorem}\} \\ \Rightarrow \exists z \in [x, y]: |d_j(x) - d_j(y)| \\ &\leq \|\partial_x \chi(t_j, z)\| \|x - y\| = \lambda_z \cdot \|x - y\| \end{aligned}$$

D35:

$$\begin{aligned} \Pr(|d_j(x) - d_j(y)| \leq \\ \leq (\lambda_x + \Delta\lambda_{x,\mathcal{V}} \cdot \|x - y\|) \cdot \|x - y\|) &\geq 1 - \gamma \end{aligned}$$

D36:

$$\begin{aligned} \Pr(|d_j(x) - d_j(y)| \leq \\ \leq (\lambda_x + \Delta\lambda_{x,\mathcal{V}} \cdot r_x) \cdot r_x) &\geq 1 - \gamma \end{aligned} \quad (\text{S28})$$

$$\begin{aligned} \text{D37:} \quad & \Pr(|d_j(x) - d_j(y)| \leq \mu \cdot \bar{m}_{j,\mathcal{V}} - d_j(x)) \geq \\ & \geq 1 - \gamma \quad \forall y \in B(x, r_x)^S \end{aligned} \quad (\text{S29})$$

$$\text{D38:} \quad \Pr(d_j(y) - d_j(x) \leq \mu \cdot \bar{m}_{j,\mathcal{V}} - d_j(x)) \geq 1 - \gamma$$

$$\text{D39:} \quad \Longleftrightarrow$$

$$\text{D40:} \quad \Pr(d_j(y) \leq \mu \cdot \bar{m}_{j,\mathcal{V}}) \geq 1 - \gamma, \quad (\text{S30})$$

$$\text{D41:} \quad \forall \gamma \in (0, 1), \exists N \in \mathbb{N} \text{ s.t. } \Pr(\mu \cdot \bar{m}_{j,\mathcal{V}} \geq m_j^*) \geq 1 - \gamma \quad (\text{S31})$$

$$\text{D42:} \quad p_{r_x} = \Pr(B(x, r_x)^S \ni x_j^*) = \frac{\text{Area}(B(x, r_x)^S)}{\text{Area}(\mathcal{B}_0)} \quad (\text{S32})$$

$$\text{D43:} \quad \Pr(\exists y \in \mathcal{V}: B(y, r_y)^S \ni x_j^*) = 1 - \prod_{x \in \mathcal{V}} (1 - p_{r_x}) \quad (\text{S33})$$

$$\text{D44:} \quad \mu \cdot \bar{m}_{j,\mathcal{V}} - d_j(x) \quad (\text{S34})$$

$$\text{D45:} \quad \geq \mu \cdot \bar{m}_{j,\mathcal{V}} - \bar{m}_{j,\mathcal{V}} = (\mu - 1) \cdot \bar{m}_{j,\mathcal{V}} \quad (\text{S35})$$

$$\text{D46:} \quad \geq (\mu - 1) \cdot d_j(x_{j,1}), \quad (\text{S36})$$

$$\begin{aligned} \text{D47:} \quad & r_{\text{bound}} = \\ & = \frac{-\lambda_x + \sqrt{\lambda_x^2 + 4 \cdot \Delta \lambda_{x,\mathcal{V}} \cdot (\mu - 1) \cdot d_j(x_{j,1})}}{2 \cdot \Delta \lambda_{x,\mathcal{V}}} \leq \\ & \leq r_x \quad \forall x \in \mathcal{V} \end{aligned}$$

$$\begin{aligned} \text{D48:} \quad & \Rightarrow \Pr(\exists y \in \mathcal{V}: B(y, r_y)^S \ni x_j^*) \geq \\ & \geq 1 - (1 - p_{r_{\text{bound}}})^N \end{aligned} \quad (\text{S37})$$

$$\text{D49:} \quad \Pr(d_j(y) \leq \mu \cdot \bar{m}_{j,\mathcal{V}}) \geq 1 - \hat{\gamma} = \sqrt{1 - \gamma} \quad (\text{S38})$$

$$\text{D50:} \quad \Pr(d_j(x^*) \leq \mu \cdot \bar{m}_{j,\mathcal{V}} | \exists x \in \mathcal{V}: B(x, r_x)^S \ni x^*) \geq \sqrt{1 - \gamma}$$

$$\text{D51:} \quad A = (\mu \cdot \bar{m}_{j,\mathcal{V}} \geq m_j^*) \quad (\text{S39})$$

$$\text{D52:} \quad B = (\exists x \in \mathcal{V}: B(x, r_x)^S \ni x_j^*) \quad (\text{S40})$$

Inline

- I1: 1^*
- I2: 2
- I3: 3
- I4: 4
- I5: 1
- I6: x_1
- I7: x_2
- I8: x_0
- I9: $\mathcal{B}_0 = B(x_0, \delta_0)$
- I10: δ_0
- I11: x
- I12: $\mathcal{B}_0$
- I13: $\chi(t_j, x_0)$
- I14: $\chi(t_j, x)$
- I15: $d_j(x)$
- I16: δ_j
- I17: $\mathcal{B}_j$
- I18: μ
- I19: $1 - \gamma$
- I20: t_0
- I21: r_x
- I22: y
- I23: $\mathfrak{B}_0$
- I24: $\partial_t x = f(x)$
- I25: t_j
- I26: $\chi(t_j, y)$
- I27: $d_j(y)$
- I28: $\mathfrak{B}_j$
- I29: $\geq 1 - \gamma$
- I30: $\delta_j = \mu \max_{x \in \mathfrak{B}_0} d_j(x)$
- I31: x_x
- I32: x_y
- I33: x_z
- I34: $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$
- I35: $\delta_t x = 1$
- I36: $x \in \mathcal{B}_0$
- I37: k
- I38: T
- I39: $t_0 < \dots < t_k = T$
- I40: $x(t_0) = x$
- I41: $x \in \mathcal{V}$
- I42: $\mathcal{B}_j = B(\chi(t_j, x_0), \delta_j(\mathcal{B}_0))$
- I43: $\mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_k$
- I44: $d_j(x) = d(t_j, x)$
- I45: $x \mapsto \chi(t_j, x)$
- I46: $\chi(t_j, \mathcal{B}_0)$
- I47: $\mathcal{B}_0^S = \text{surface}(\mathcal{B}_0)$
- I48: $t_0 < \dots < t_k = T$
- I49: $\mu > 1$
- I50: $\gamma \in (0, 1)$
- I51: b
- I52: d
- I53: $\mathcal{V} \leftarrow \{\}$
- I54:
- I55: $x^B \in \mathcal{B}_0^S$

I56: $(j = 1; j \leq k; j = j + 1)$
 I57: $\bar{p} \leftarrow 0$
 I58: $\bar{p} < 1 - \gamma$
 I59: $\mathcal{V} \leftarrow \mathcal{V} \cup \{x^B\}$
 I60: $x_j \leftarrow \chi(t_j, x_0)$
 I61: $\bar{m}_{j,\mathcal{V}} \leftarrow \max_{x \in \mathcal{V}} d(t_j, x)$
 I62: λ_x
 I63: $x \in \mathcal{V}$
 I64: $\Delta\lambda_{x,\mathcal{V}}$
 I65: $r_x(\lambda_x, \Delta\lambda_{x,\mathcal{V}})$
 I66: $\mathcal{S} \leftarrow \bigcup_{x \in \mathcal{V}} B(x, r_x)^S$
 I67: $\bar{p} \leftarrow \Pr(\mu \cdot \bar{m}_{j,\mathcal{V}} \geq m^*)$
 I68: $x^B \in \mathcal{B}_0$
 I69: $\delta_j \leftarrow \mu \cdot \bar{m}_{j,\mathcal{V}}$
 I70: $\mathcal{B}_j \leftarrow B(x_j, \delta_j)$
 I71: $(\mathcal{B}_1, \dots, \mathcal{B}_k)$
 I72: t_j ($t_0 < \dots < t_k = T$)
 I73: $x^B \in \mathcal{B}_0^S$
 I74: $\bar{m}_{j,\mathcal{V}}$
 I75: $\mathcal{V}$
 I76: $\bar{m}_{j,\mathcal{V}} = \max_{x \in \mathcal{V}} d_j(x)$
 I77: $B(x, r_x)^S = B(x, r_x) \cap \mathcal{B}_0^S$
 I78: $B(x, r_x)^S$
 I79: γ, t_j
 I80: $\Pr(d_j(y) \leq \mu \cdot \bar{m}_{j,\mathcal{V}}) \geq 1 - \gamma$
 I81: $y \in B(x, r_x)^S$
 I82: $m_j^* = d_j(x_j^*) = \max_{x \in \mathcal{B}_0} d_j(x)$
 I83: $\mu \cdot \bar{m}_{j,\mathcal{V}}$
 I84: γ
 I85: $B(x, r_x)^S$
 I86: f
 I87: $N = |\mathcal{V}|$
 I88: $\chi(t_j, x)$
 I89: $\lambda_x = \|\partial_x \chi(t_j, x)\|$
 I90: $\hat{\gamma} = 1 - \sqrt{1 - \gamma}$
 I91: $\Delta\lambda_{\mathcal{V}}$
 I92: $\sqrt{1 - \gamma}$
 I93: $F_{L,\hat{\gamma}}$
 I94: γ, t_j
 I95: $\max_{x,y} |\lambda_x - \lambda_y|/\|x - y\|$
 I96: $\Pr(\lambda_y \leq \lambda_x + \Delta\lambda_{x,\mathcal{V}} \cdot \|x - y\|) \geq 1 - \gamma$
 I97: $(\mu \cdot \bar{m}_{j,\mathcal{V}} - d_j(x)) = \lambda_x r_x + \Delta\lambda_{x,\mathcal{V}} r_x^2$
 I98: $m_j^* = \max_{x \in \mathcal{B}_0} d_j(x)$
 I99: $N = |\mathcal{V}|$
 I100: x_j^*
 I101: $d_j(x_j^*) = m_j^*$
 I102: $\gamma \in (0, 1)$
 I103: $\exists N \in \mathbb{N}: \Pr(\exists x \in \mathcal{V}: B(x, r_x)^S \ni x_j^*) \geq \sqrt{1 - \gamma}$
 I104: N
 I105: $1 - \sqrt{1 - \gamma}$
 I106: $\Pr(d_j(x^*) \leq \mu \cdot \bar{m}_{j,\mathcal{V}} | \exists x \in \mathcal{V}: B(x, r_x)^S \ni x^*) \geq \sqrt{1 - \gamma}$
 I107: A, B
 I108: $\Pr(A) \geq \Pr(A \cap B) = \Pr(A|B) \cdot \Pr(B)$
 I109: $\mu = 1.1$
 I110: $1 - \lambda$
 I111: $t \approx 20s$

I112: $t > 40s$
 I113: $\dot{x} = Ax + Bu$
 I114: u
 I115: λ
 I116: $F_{L,\gamma}$
 I117: m
 I118: $(a_1, \dots, a_m)$
 I119: $(b_1, \dots, b_m)$
 I120: $g : \mathbb{R}^n \rightarrow \mathbb{R}$
 I121: $X = \max_{i=1}^m |g(a_i) - g(b_i)| / \|a_i - b_i\|$
 I122: F
 I123: $(x_1, \dots, x_n) \sim X$
 I124: $\hat{F}_n(x) = \sum_{i=1}^n \mathbb{1}_{x_i \leq x}$
 I125: G_n
 I126: $\hat{F}_n$
 I127: D_n^-
 I128: $\alpha = \min(\gamma, 0.5)$
 I129: $\epsilon_{n,\gamma}$
 I130: $\epsilon \geq \sqrt{\frac{1}{2n} \ln 2}$
 I131: $\gamma = e^{-2n\epsilon^2}$
 I132: ϵ
 I133: $x(t_0) \in B(x_0, \delta_0)$
 I134: $\{x_1, \dots, x_n\}$
 I135: n
 I136: $X = \max_{i=1}^m |\lambda_{a_i} - \lambda_{b_i}| / \|a_i - b_i\|$
 I137: X
 I138: $Pr(X \leq \Delta\lambda) = F(\Delta\lambda)$
 I139: $F_{L,\hat{\gamma}(x)} \leq F(x)$
 I140: $F_{L,\hat{\gamma}}(\Delta\lambda)$
 I141: $Y = |\lambda_a - \lambda_b| / \|a - b\|$
 I142: a, b
 I143: $x, y \in \mathcal{V}$
 I144: $\lambda_x + \Delta\lambda_{x,\mathcal{V}} \cdot \|x - y\|$
 I145: λ_z
 I146: $\|x - y\| \leq r_x$
 I147: $\mu \cdot \bar{m}_{j,\mathcal{V}} - d_j(x) = \lambda_x r_x + \Delta\lambda_{x,\mathcal{V}} r_x^2$
 I148: $d_j(y) \leq d_j(x)$
 I149: $d_j(y) \geq d_j(x)$
 I150: $d_j(y) \leq d_j(x) \leq \mu \cdot \bar{m}_{j,\mathcal{V}}$
 I151: $x_{j,1}$
 I152: $N \rightarrow \infty$
 I153: $\forall \gamma \in (0, 1) \exists N \in \mathbb{N}: \Pr(\exists x \in \mathcal{V}: B(x, r_x)^S \ni x_j^*) \geq \sqrt{1 - \gamma}$
 I154: $\Delta\lambda_x$
 I155: $\forall y \in B(x, r_x)^S$
 I156: $B(x, r_x)^S \ni x_j^*$
 I157: x_j^*
 I158: $\Pr(\mu \cdot \bar{m}_{j,\mathcal{V}} \geq m_j^*) \geq \Pr(A|B) \cdot \Pr(B) = 1 - \gamma$